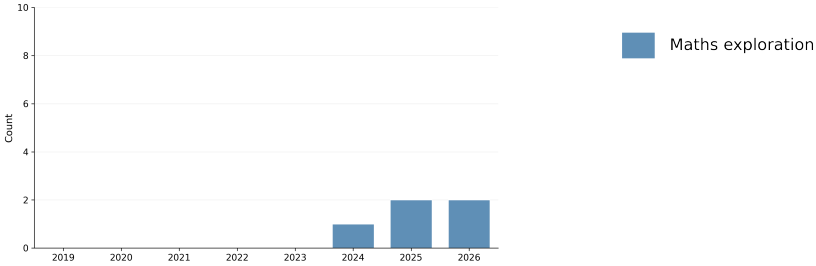
From Physics-informed Neural Networks to Numerically Verified Proofs in Geometric Analysis

Daniel Platt (Imperial College London)
Isaac Newton Institute, 21 Apr 2026

There are many elliptic equations in geometric analysis for which it is not known whether a solution exists. In many cases, good numerical solutions using PINN and other methods are known. There are only few numerically verified proofs in geometry, even though the equations are simple from an analysis point of view. In the talk I will mention the limited scenarios in which numerically verified proofs succeeded and I will point out the two places where geometry makes proving spectral gaps and interior estimates difficult, even for the simplest differential operators.

Exploration and numerically verified proofs



- Numerically verified proofs only in special cases
  I.e.: S^2 or symmetry reduction to $\mathbb{R}$ or $\mathbb{R}^2$
- Open questions with good numerics:
  Does a solution exist?
  (E.g. minimal surfaces [Sch21])
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- Physics: (Twisted) Laplace spectra

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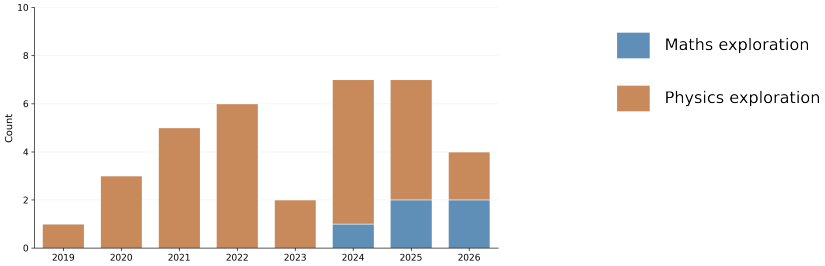

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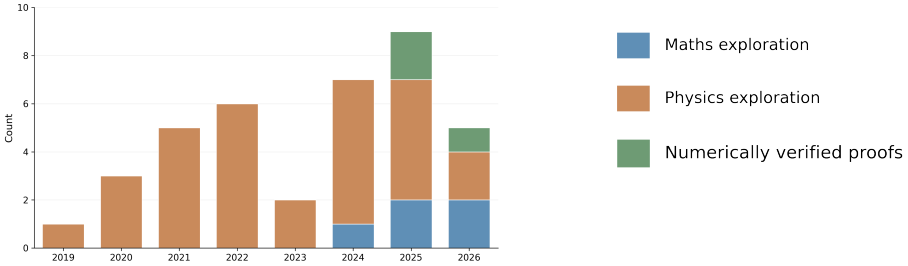

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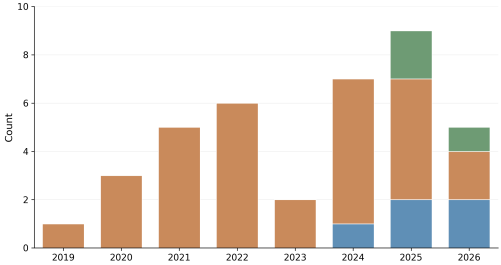

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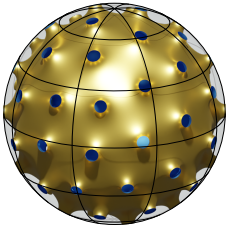

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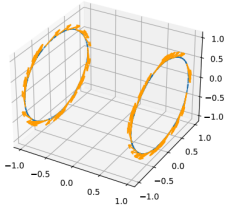

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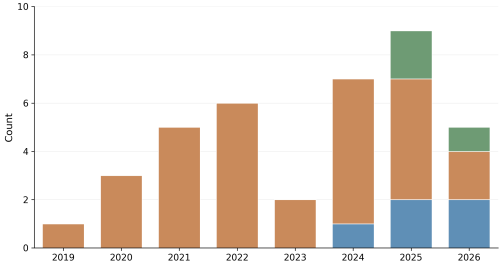

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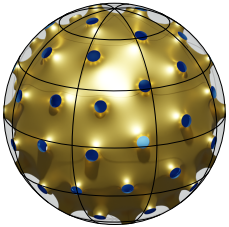

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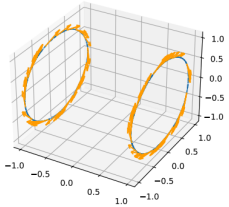

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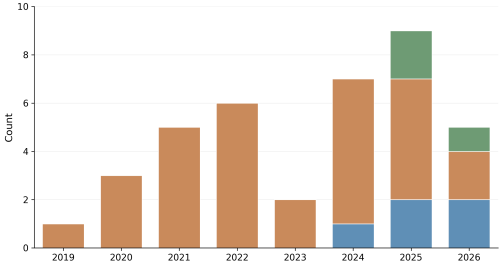

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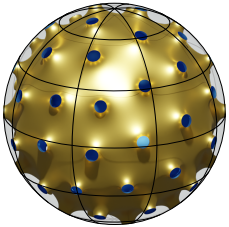

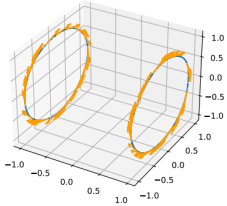

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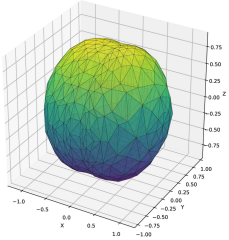
Example: Nirenberg problem (PINN)

- Question: which $K : S^2 \to \mathbb{R}$ are curvatures of some metric?



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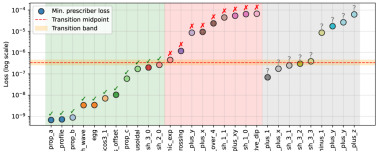

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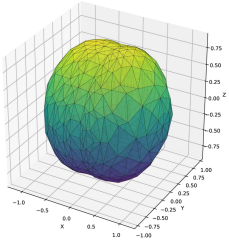

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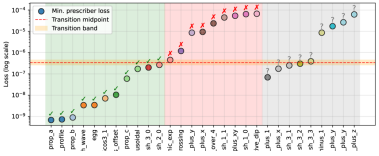

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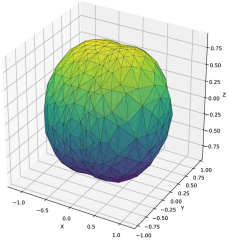

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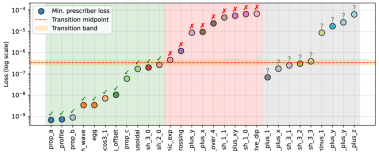

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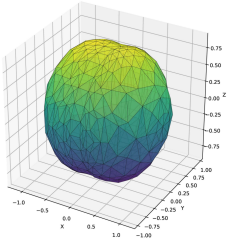

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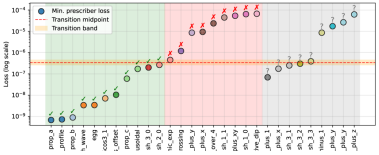

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- Numerically verified proof for elliptic PDE
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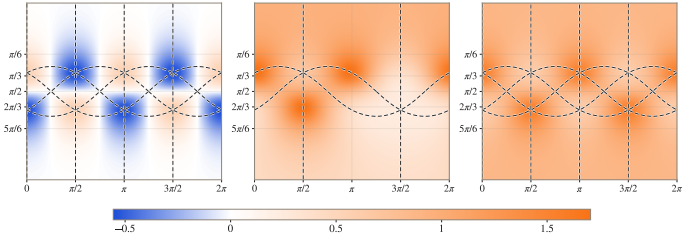

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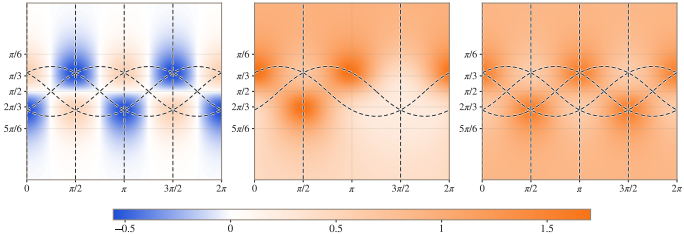

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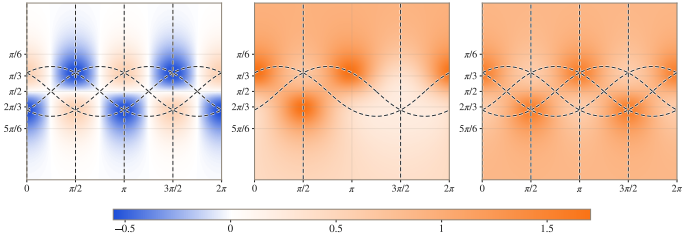

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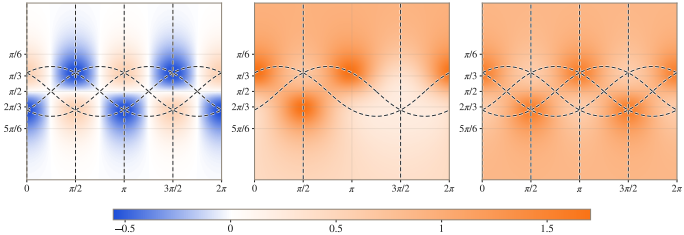

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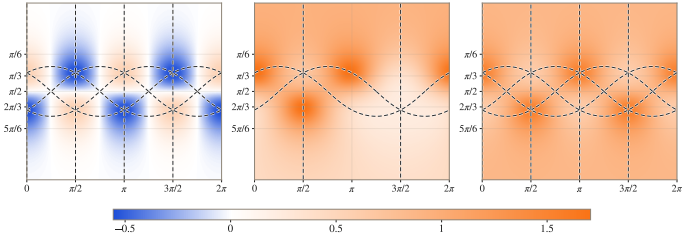

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Thank you for the attention!

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- General purpose theorems, e.g.:
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  - Cheeger $\lambda_1(\Delta) \geq \{\text{Cheeger const.}\}$
- On nice spaces use (pseudo)-spectral basis [Pla26, RT19], cf. J. Henkel's talk
- Symmetries $\rightsquigarrow$ ODE $\rightsquigarrow$ shooting problem [BH26, Wan25]
- Discretisation (FEM, DEC) $\Rightarrow \lambda_1^h$
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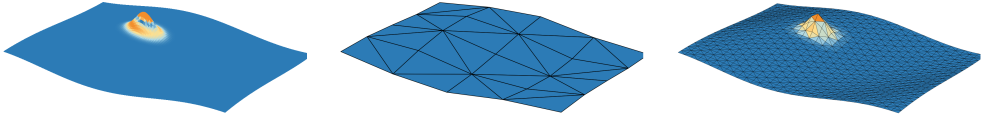

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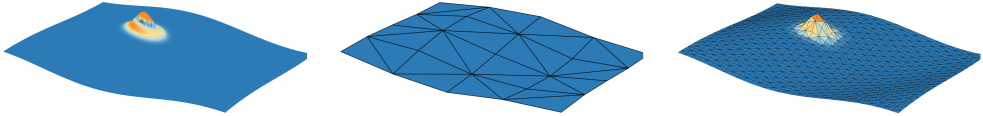

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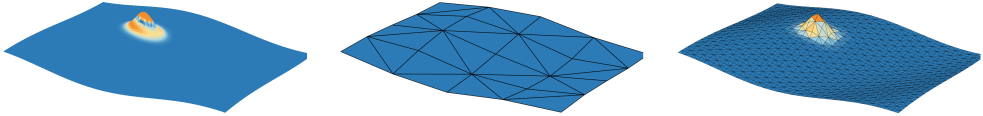

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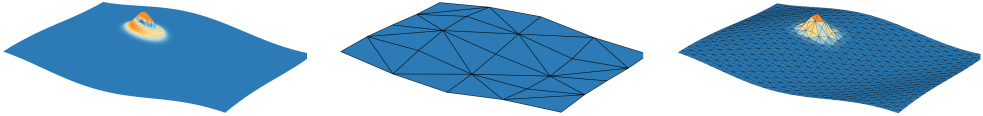

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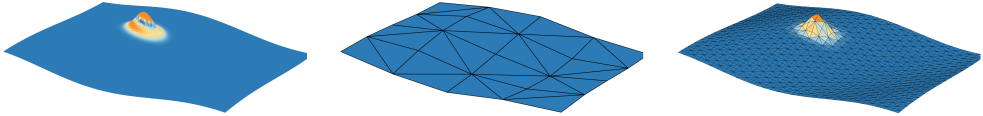

Diificulty 2: Operator estimates on manifolds

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- On surface with $|\text{curvature}| \leq K$: $\left|\left|\nabla^2 u\right|\right|_{L^2} \lesssim \left|\left|\Delta u\right|\right|_{L^2} + K \left|\left|u\right|\right|_{L^2}$
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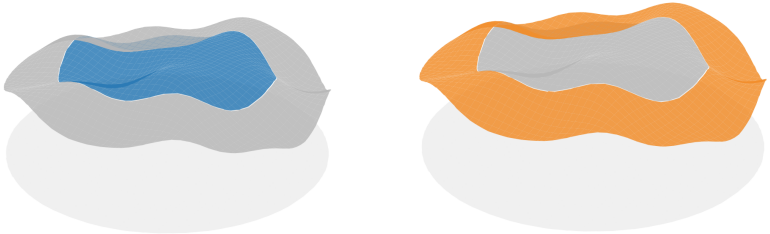

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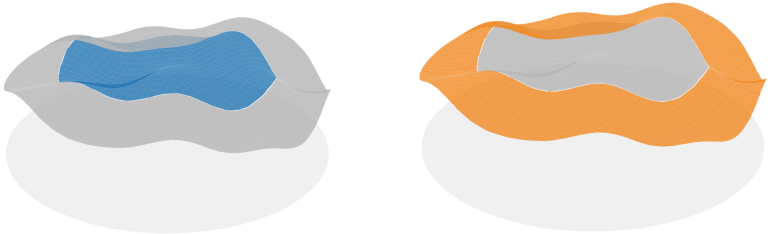

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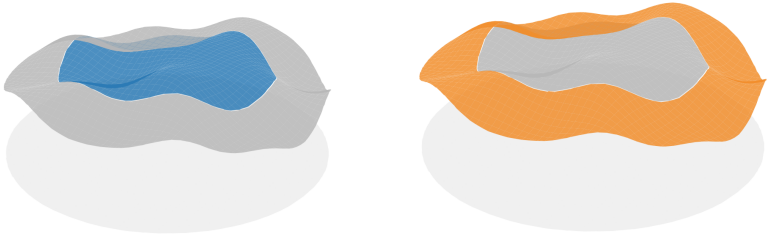
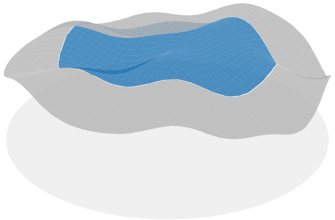
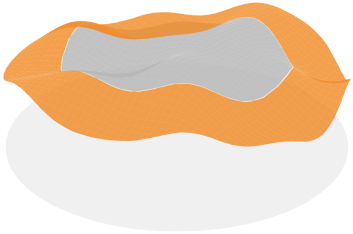


Diificulty 2: Operator estimates on manifolds

- On $\Omega \subset \mathbb{R}^2$ with Dirichlet: $\left|\left|\nabla^2 u\right|\right|_{L^2} = \left|\left|\Delta u\right|\right|_{L^2}$
- On surface with $|\text{curvature}| \leq K$: $\left|\left|\nabla^2 u\right|\right|_{L^2} \lesssim \left|\left|\Delta u\right|\right|_{L^2} + K \left|\left|u\right|\right|_{L^2}$
- $W^{k,p}$ and $C^{k,\alpha}$ dependence on curvature?
- Computational methods for these constants?





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